Experiment no:-7

Title:- Write programs to demonstrate user-defined functions, lambda functions, and recursion.
#User-defined functions

1.Write a user-defined function to find the largest of three numbers.

```
def largest(a, b, c):

if a >= b and a >= c:

    return a   elif b >=

a and b >= c:

    return b

else:

    return c print("Largest:",

largest(10, 25, 15))
```



2. Create a function to calculate the factorial of a given number.

```
def factorial(n):

  fact = 1   for i in

range(1, n + 1):

    fact *= i

return fact

print("Factorial:", factorial(5))
```



3. Define a function to check whether a given number is prime or not.

```
def is_prime(n):

  if n <= 1:
```

```
    return False    for i in
range(2, int(n/2) + 1):       if n
% i == 0:           return False
return True
print("Prime:", is_prime(7))
```



4. Write a function that takes a string as input and returns the number of vowels.

```
def count_vowels(s):
vowels = 'aeiouAEIOU'
count = 0    for ch in s:
if ch in vowels:
count += 1    return
count
print("Number of vowels:", count_vowels("Education"))
```



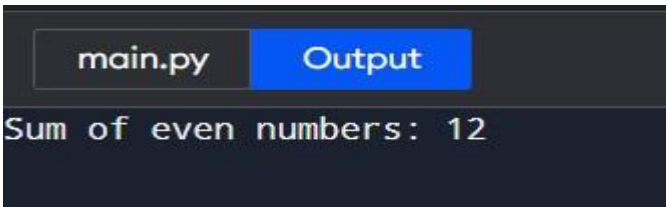
5. Create a function that converts temperature from Celsius to Fahrenheit.

```
def celsius_to_fahrenheit(c):
   return (c * 9/5) + 32 print("Fahrenheit:",
celsius_to_fahrenheit(25))
```



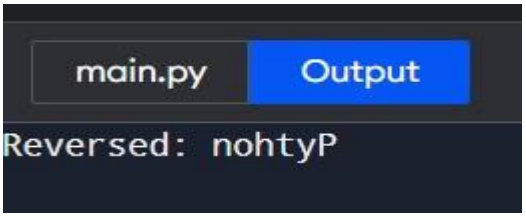
6. Define a function to find the sum of all even numbers in a list.

```
def sum_even(lst):
   return sum([x for x in lst if x % 2 == 0])
print("Sum of even numbers:", sum_even([1,2,3,4,5,6]))
```



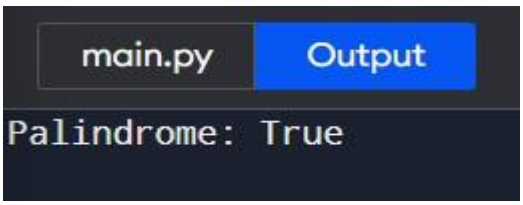
7. Write a function that reverses a given string.

```
def reverse_string(s):
    return s[::-1]
print("Reversed:", reverse_string("Python"))
```



8. Create a function that checks whether a string is a palindrome.

```
def is_palindrome(s):
return s == s[::-1]
print("Palindrome:", is_palindrome("madam"))
```



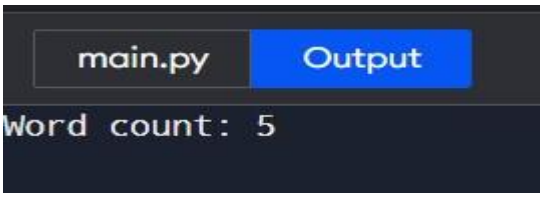
9. Define a function that returns the maximum and minimum values from a list.

```
def max_min(lst):
    return max(lst), min(lst) print("Max and
Min:", max_min([5, 1, 8, 3, 2]))
```



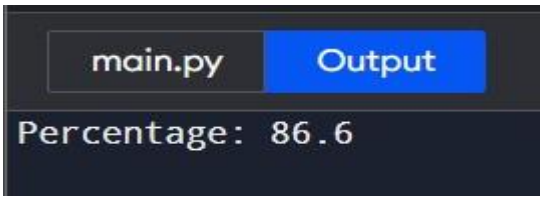
10. Write a function to count the number of words in a sentence.

```
def count_words(sentence):     words = sentence.split()     return
len(words) print("Word count:", count_words("Python is an amazing
language"))
```



11. Create a function that accepts marks in 5 subjects and returns the percentage.

```
def percentage(m1, m2, m3, m4, m5):
total = m1 + m2 + m3 + m4 + m5
   return (total / 500) * 100 print("Percentage:",
percentage(78, 85, 90, 88, 92))
```



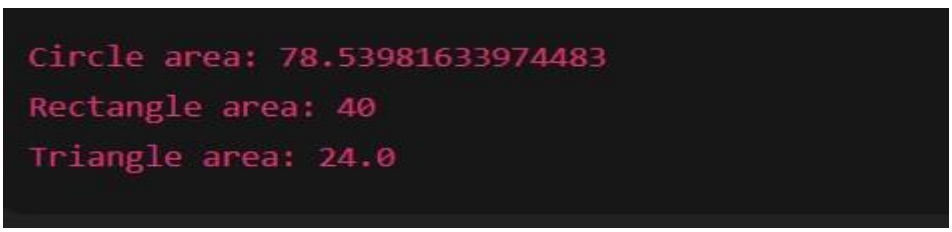
12. Define a function to calculate the area of a circle, rectangle, and triangle (use parameters). import math

```
def area_circle(r):
   return math.pi * r * r

def area_rectangle(l, b):
   return l * b

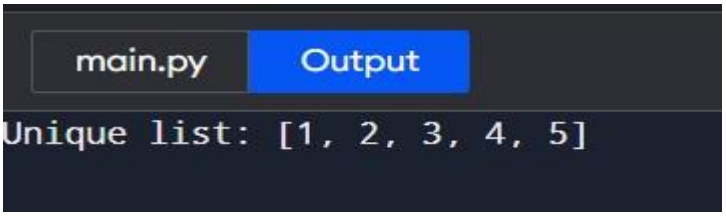
def area_triangle(base, height):
return 0.5 * base * height

print("Circle area:", area_circle(5))
print("Rectangle area:", area_rectangle(10, 4))
print("Triangle area:", area_triangle(6, 8))
```



13. Write a function that takes a list and returns a new list with only unique elements. def unique_list(lst):

return list(set(lst)) print("Unique list:", unique_list([1,2,2,3,4,4,5]))



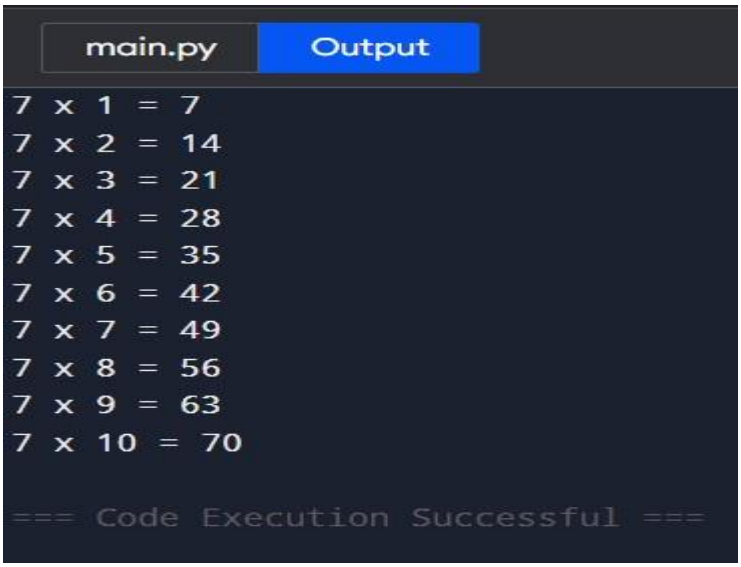
14. Create a function that accepts a number and prints its multiplication table.

def table(n): for i in

range(1, 11):

print(f"{n} x {i} = {n * i}")

table(7)

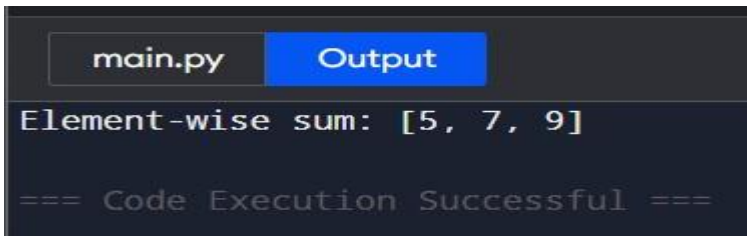


15. Define a function that takes two lists and returns their element-wise sum.

def list_sum(list1, list2):

return [a + b for a, b in zip(list1, list2)]

print("Element-wise sum:", list_sum([1,2,3], [4,5,6]))

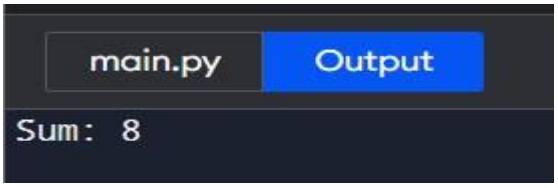


#Lambda functions

1. Write a lambda function to add two numbers.

add = lambda a, b: a + b print("Sum:",

add(5, 3))



2. Create a lambda to find the square of a number.

square = lambda x: x ** 2 print("Square:",

square(6))



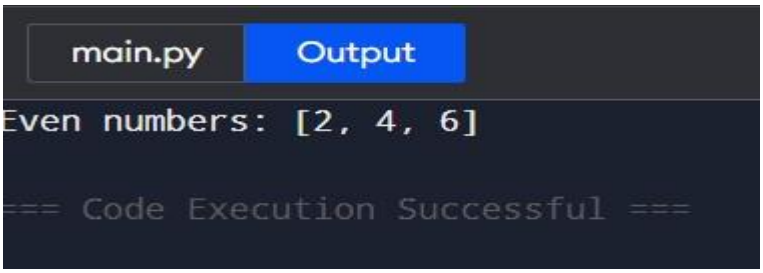
3. Write a lambda that checks whether a number is even or odd.
   check = lambda x: "Even" if x % 2 == 0 else "Odd"
   print("Number is:", check(7))



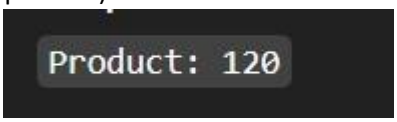
4. Use map() with lambda to double all elements in a list.
   nums = [1, 2, 3, 4, 5]
   double = list(map(lambda x: x * 2, nums)) print("Doubled list:", double)



5. Use filter() with lambda to extract only even numbers from a list.
   nums = [1, 2, 3, 4, 5, 6]
   evens = list(filter(lambda x: x % 2 == 0, nums)) print("Even numbers:", evens)



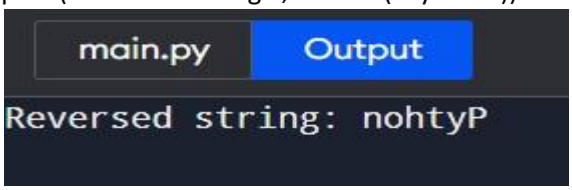
6. Use reduce() (from functools) with lambda to find the product of all numbers in a list. from functools import reduce nums = [1, 2, 3, 4, 5]
product = reduce(lambda x, y: x * y, nums) print("Product:", product)



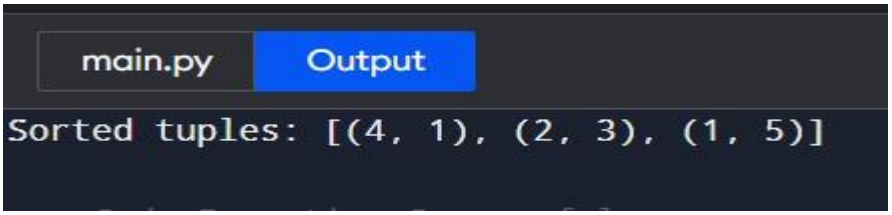
7. Create a lambda that returns the maximum of two numbers.
maximum = lambda a, b: a if a > b else b print("Maximum:", maximum(15, 25))



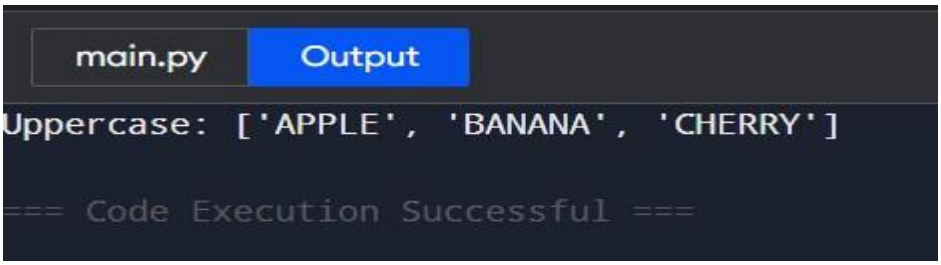
8. Write a lambda to reverse a string.
reverse = lambda s: s[::-1]
print("Reversed string:", reverse("Python"))



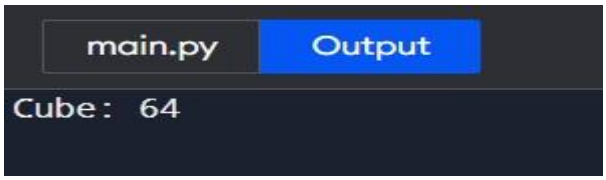
9. Use a lambda to sort a list of tuples based on the second element.
data = [(1, 5), (2, 3), (4, 1)]
sorted_data = sorted(data, key=lambda x: x[1]) print("Sorted tuples:", sorted_data)



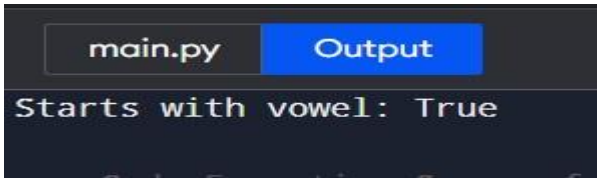
10. Write a lambda function that converts all strings in a list to uppercase.
words = ["apple", "banana", "cherry"] uppercase = list(map(lambda s: s.upper(), words))
print("Uppercase:", uppercase)



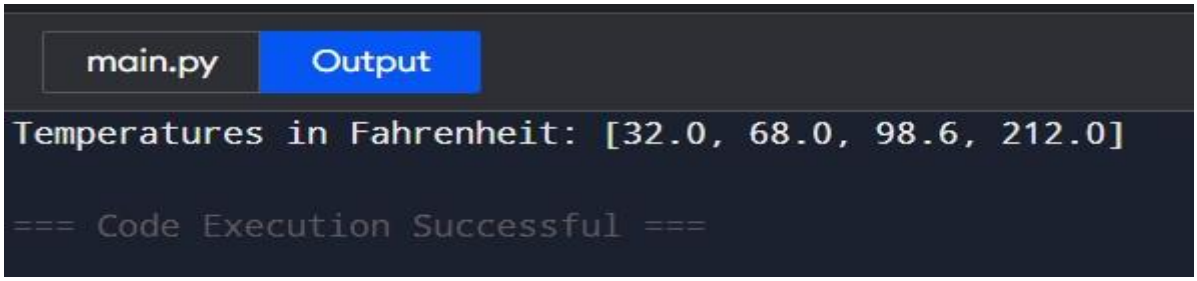
11. Create a lambda to find the cube of a number.
    cube = lambda x: x ** 3 print("Cube:",
    cube(4))



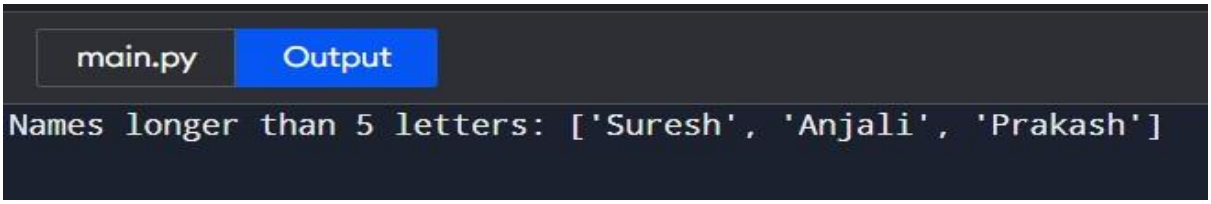
12. Write a lambda to check whether a string starts with a vowel. starts_with_vowel = lambda s:
    s[0].lower() in 'aeiou' print("Starts with vowel:", starts_with_vowel("Apple"))



13. Use map() and lambda to convert a list of temperatures from Celsius to Fahrenheit.
    temps_c = [0, 20, 37, 100]
    temps_f = list(map(lambda c: (c * 9/5) + 32, temps_c)) print("Temperatures
    in Fahrenheit:", temps_f)



14. Use filter() and lambda to find names longer than 5 characters in a list.
    names = ["Amit", "Suresh", "Anjali", "Raj", "Prakash"] long_names
    = list(filter(lambda n: len(n) > 5, names)) print("Names longer
    than 5 letters:", long_names)



15. Use a lambda and map() to calculate the square root of numbers in a list.
    import math

    nums = [4, 9, 16, 25]
    square_roots = list(map(lambda x: math.sqrt(x), nums)) print("Square
    roots:", square_roots)


main.py
Output
Square roots: [2.0, 3.0, 4.0, 5.0]